

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.***Session 2026-2027**

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

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Sessi on	2026-27 (ODD)	Course Name	Lab : Embeded Systems
Seme ster	7	Course Code	23IOT1706
Roll No	61	Name of Student	Sheikh Wasimuddin

Practical Number	04
Course Outcome	CO1: Explain the fundamentals of embedded microcontrollers, development environments, and simulation tools. CO2: Develop embedded applications using Arduino/ESP32 and interface sensors, actuators, and peripherals. CO3: Implement event-driven and communication-based embedded systems using interrupts, UART, and I2C protocols. CO4: Design, simulate, and test embedded system applications for real-world applications.
Aim	To establish UART communication between an ESP32 and an Arduino and exchange data between them.
Theory (100 words)	UART (Universal Asynchronous Receiver/Transmitter) is a communication protocol used for transferring digital data between electronic devices in a serial manner. Unlike parallel communication, UART sends data sequentially, one bit at a time. UART communication generally uses two signal lines: TX (Transmit) for sending data and RX (Receive) for receiving data. The transmitting device's TX line is connected to the receiving device's RX line, and vice versa. A common GND is required between the communicating devices. UART is an asynchronous communication protocol, meaning that it does not use a common clock line. Instead, both devices must be configured with the same baud rate, such as 9600 or 115200 bps. Each transmitted character is organized into a frame containing a start bit, data bits, optional parity bit, and stop bit. In this experiment, ESP32 and Arduino Uno are interfaced using UART to demonstrate serial data exchange. One microcontroller can transmit information while the other receives and processes it. Communication can also be made bidirectional, allowing both devices to exchange information. UART is widely used in embedded systems, sensor interfacing, GPS modules,



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	Bluetooth modules, debugging, and communication between microcontrollers because of its simple hardware requirements and easy implementation.
Procedur e and Executio n (100 Words)	1. Connect the ESP32 and Arduino Uno using suitable jumper wires. 2. Connect the TX (Transmit) pin of the ESP32 to the RX (Receive) pin of the Arduino Uno. 3. Connect the TX pin of the Arduino Uno to the RX pin of the ESP32. 4. Connect the GND pins of both boards together to establish a common ground. 5. Launch the Arduino IDE and select the appropriate board and COM port. 6. Create and write the UART communication programs for the ESP32 and Arduino Uno. 7. Configure both boards with the same baud rate, such as 9600 bps. 8. Upload the ESP32 program to the ESP32 board. 9. Upload the Arduino program to the Arduino Uno. 10. Open the Serial Monitor and select the same baud rate used in the programs. 11. Transmit a test message from the ESP32 and observe it on the Arduino Uno. 12. Transmit a response message from the Arduino Uno and verify it on the ESP32. 13. Check the received messages for correct data transmission without errors. 14. Repeat the communication test with different messages or values. 15. Confirm that two-way UART communication is working successfully between the ESP32 and Arduino Uno CODE: ESP32 //sheikh wasimuddin es lab 4 cseiot 61 #include <Arduino.h>



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```
#include <HardwareSerial.h>

// UART2 on ESP32
HardwareSerial Serial2(2);

// ESP32 UART pins
#define TXD2 17
#define RXD2 16

void setup() {
    // Serial Monitor
    Serial.begin(9600);

    // UART2 communication
    Serial2.begin(9600, SERIAL_8N1, RXD2, TXD2);

    delay(1000);

    Serial.println("ESP32 UART Communication Started");
    Serial.println("TX2 = GPIO17, RX2 = GPIO16");

    // Send message to Arduino Uno
    Serial2.println("Hello from ESP32");
}

void loop() {

    // Check if data is received from Arduino Uno
    if (Serial2.available()) {
        String data = Serial2.readStringUntil('\n');

        Serial.print("Received from Arduino: ");
        Serial.println(data);
    }

    // Send message to Arduino every 5 seconds
    Serial2.println("Message from ESP32");
}
```



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```
Serial.println("Sent: Message from ESP32");
```

```
delay(5000);  
}
```

CODE UNO:

```
//sheikh wasimuddin es lab 4 cseiot 61
```

```
void setup() {  
  Serial.begin(9600);
```

```
  Serial.println("Arduino Uno UART Communication  
Started");  
}
```

```
void loop() {
```

```
  // Receive data from ESP32  
  if (Serial.available()) {  
    String data = Serial.readStringUntil('\n');
```

```
    Serial.print("Received from ESP32: ");  
    Serial.println(data);
```

```
    // Send response back to ESP32  
    Serial.println("Hello from Arduino Uno");  
  }  
}
```



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Stepwise Screenshots with steps:

```
1 //sheikh wasimuddin es lab 4 cseiot 61
2 #include <Arduino.h>
3 #include <HardwareSerial.h>
4 HardwareSerial Serial2(2); // Use UART2 on ESP32 (TX2=GPIO17, RX2=GPIO16)
5
6 void setup() {
7     Serial.begin(9600); // For Serial Monitor (optional)
8     Serial2.begin(9600, SERIAL_8N1); // Initialize UART2 with 9600 baud
9     delay(1000);
10    Serial.println("ESP32: UART Communication Started (TX2=17, RX2=16)");
11    Serial2.println("Hello from ESP32"); // Send message to Arduino Uno
12    delay(2000);
13 }
14
15 void loop() {
16     if (Serial2.available()) { // Check if data is received from Arduino
17         String data = Serial2.readStringUntil('\n');
18         Serial.print("Received from Arduino: ");
19         Serial.println(data);
20     }
21     // Send a message to Arduino Uno every 5 seconds
22     Serial2.println("Message from ESP32");
23     delay(5000);
24 }
```



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```
Arduino Uno Code | Arduino IDE 2.3.6
File Edit Sketch Tools Help

[Checkmark] [Next] [Previous] [USB] Arduino Uno

Arduino_Uno_Code.ino x
1 //sheikh wasimuddin es lab 4 cseiot 61
2
3 void setup() {
4     Serial.begin(9600);
5     Serial.println("Arduino Uno UART Communication Started");
6 }
7
8 void loop() {
9     // Receive data from ESP32
10    if (Serial.available()) {
11        String data = Serial.readStringUntil('\n');
12        Serial.print("Received from ESP32: ");
13        Serial.println(data);
14        // Send response back to ESP32
15        Serial.println("Hello from Arduino Uno");
16    }
17 }
```




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Conclusi on	The servo motor was successfully interfaced with the Arduino Uno and controlled using the Servo library . The motor was programmed to rotate to 0°, 90°, and 180° at specified time intervals. The practical demonstrated how Arduino can be used to control the precise angular position of a servo motor. It also provided an understanding of servo interfacing, signal control, and PWM-based operation , which are useful in robotics, automation, and embedded systems applications.
Date	20-08-2026